

Francesco Piatti

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Education

Imperial College London – University of Oxford

PHD, MATHEMATICS

London – Oxford, UK

Sept 2023 – Present

- EPSRC studentship in the Centre for Doctoral Training (CDT) in Mathematics of Random Systems.
- Supervised by Prof. Thomas Cass (Professor of Mathematical Finance).
- *Research interests*: Neural SDEs, State Space Models, Rough Paths and Signatures, AI for Finance, Algorithmic Trading.
- *Teaching assistant*: Interest Rate Models (2026, 2025), Statistics for Finance (2026, 2025), Applied Trading Strategies – Imperial Business School (2026, 2025), Probability for Statistics (2024, 2023), Mathematics for ML (2024).

Working Papers:

- Deep Calibration of Interest Rate Models with Neural SDEs. Joint with Prof. Brigo and Dr. Ferrucci.
- The Free Signature Transform and Schwinger-Dyson Kernels. Joint with Prof. Cass and Dr. Bizzi.
- Cubature on Wigner Space. Joint with Prof. Cass and Dr. Bizzi.

Imperial College London

MSci, MATHEMATICS – FIRST CLASS HONOURS, DEAN'S LIST FINAL YEAR

London, UK

Sept 2019 – June 2023

- Thesis: “Deep Hedging with Signatures and Neural Rough Differential Equations” – achieved First Class.

Liceo Scientifico Filippo Lussana

ITALIAN SCIENTIFIC HIGH SCHOOL DIPLOMA – OVERALL GRADE: 100/100

Bergamo, Italy

Sept 2014 – July 2019

- Standardised tests: SAT Subject test – Math Level 1: 800/800; SAT Subject test – Math Level 2: 800/800.

Publications and Pre-Prints

Piatti, F., Iannucci, A., & Cass, T. (2026). Structured Neural SDEs for Functional Calibration. *Under review at ICLR 2027*.

Ferrera, A., Piatti, F., & Cass, T. (2026). Propagating Log-Signatures: Lie-Algebraic Models for Sequential Data. *Pre-Print*.

Piatti, F., Cass, T., & Turner, W. (2026). Random Controlled Differential Equations. In *ICLR 2026*, pp. 84345–84374.

Cass, T., Piatti, F., & Pei, J. (2025). Numerical Schemes for Signature Kernels. *SIAM Journal on Numerical Analysis*, 63(6), pp.2371–2394.

Calvo-Ordóñez, S*, Meunier, M*, Piatti, F*, & Shi, Y* (2024). Partially Stochastic Infinitely Deep Bayesian Neural Networks. In *ICML 2024*, pp.5436–5452, *PMLR*. (*Equal contributions).

Work Experience

G-Research

SPRING INTERN

London, UK

April 2025

- Participated in advanced workshops on Mathematics, Statistics, Quantitative Finance, and Algorithmic Trading.

Department of Mathematical Finance, Imperial College London

STUDENT RESEARCHER, SUPERVISED BY PROF. THOMAS CASS

London, UK

July – Sept 2021, July – Sept 2022

- Worked on US recession forecasting using machine learning techniques including neural networks, kernel methods and signature transforms.
- Improved accuracy of state-of-the-art models presented in the Federal Reserve paper: Puglia, M., & Tucker, A. (2020). *Machine learning, the treasury yield curve and recession forecasting*.
- Developed a novel method leveraging signatures to estimate the correlation of multivariate time series with non-synchronous observations.
- Worked efficiently in a team of 3 people to design and implement statistical tests and simulation pipelines to evaluate model robustness.

Studio Conca Jannone Maffei

SUMMER INTERN

Bergamo, Italy

June – July 2020

- Conducted financial analysis for clients and assisted in developing management and investment strategies during the COVID-19 recovery phase.
- Contributed to the development of a Python-based software to automate the analysis of financial statements and balance sheets.

Additional Information

Languages Italian (native), English (fluent), Spanish (intermediate).

- Interests and Personal Projects**
- Built *ReadingRoom*, an open-source desktop app for cataloguing papers with AI digests and citation graphs.
 - Systematic Trading – building a Python backtesting library with LLM integration to assist strategy research.

Programming Advanced programming in Python (incl. PyTorch, JAX, pandas), intermediate in R. HPC projects on computer clusters (CPU/GPU multi-threading and multi-processing). Advanced knowledge of Office 365 and LaTeX.

- Positions of Responsibility**
- Member of board of directors and board of executives of United Italian Societies Ltd (June 2021 – June 2023).
 - President of Imperial College Italian Society (June 2021 – June 2023).